

Judge0 Installation and Troubleshooting Report

Summary

This document records the troubleshooting process, root cause analysis, configuration changes, and final successful validation of Judge0 on Ubuntu 24.04.

Detailed Log

You have confirmed:

```
cat /proc/cmdline
BOOT_IMAGE=/boot/vmlinuz-6.17.0-35-generic root=... ro quiet splash
```

```
stat -fc %T /sys/fs/cgroup
cgroup2fs
```

This means cgroup v1 is NOT enabled yet.

Step 1: Edit GRUB

Open the GRUB configuration:

```
sudo nano /etc/default/grub
```

Find this line:

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash"
```

Replace it with:

```
GRUB_CMDLINE_LINUX_DEFAULT="quiet splash systemd.unified_cgroup_hierarchy=0
cgroup_enable=memory cgroup_memory=1 swapaccount=1"
```

Save:

```
CTRL + O
ENTER
CTRL + X
Step 2: Update GRUB
```

Run:

```
sudo update-grub
```

You should see output similar to:

```
Generating grub configuration file ...
Found linux image ...
done
Step 3: Reboot
sudo reboot
Step 4: Verify After Reboot
```

Login again and run:

```
cat /proc/cmdline
```

You should now see:

```
systemd.unified_cgroup_hierarchy=0
cgroup_enable=memory
cgroup_memory=1
swapaccount=1
```

Then run:

```
stat -fc %T /sys/fs/cgroup
If you get:
tmpfs
```

■ Success — cgroup v1 is enabled.

If you still get:
cgroup2fs

■ Ubuntu 24.04 kernel is forcing cgroup v2 and Judge0 v1.13.1 will not work properly on this OS.

Step 5: Restart Judge0

If cgroup v1 is enabled:

```
cd ~/judge0-v1.13.1
```

```
sudo docker compose down -v
```

```
sudo docker compose up -d
```

Step 6: Test Judge0

```
curl -X POST http://localhost:2358/submissions/?wait=true \
-H "Content-Type: application/json" \
-d '{
  "source_code": "print(\"Hello\")",
  "language_id": 71
}'
```

Expected:

```
{
  "stdout": "Hello\n",
  "status": {
    "id": 3,
    "description": "Accepted"
  }
}
```

Important

After you run:

```
sudo update-grub
```

sudo reboot

send me the output of:

```
cat /proc/cmdline
stat -fc %T /sys/fs/cgroup
```

and I'll tell you the next step.

```
infyshine@infyshine:~$ cat /proc/cmdline
BOOT_IMAGE=/boot/vmlinuz-6.17.0-35-generic
root=UUID=18249670-7e5e-4047-980b-a440a0c75e76 ro quiet splash
systemd.unified_cgroup_hierarchy=0 cgroup_enable=memory cgroup_memory=1 swapaccount=1
infyshine@infyshine:~$ stat -fc %T /sys/fs/cgroup
tmpfs
infyshine@infyshine:~$ sudo docker compose down -v
no configuration file provided: not found
infyshine@infyshine:~$ cd ~/judge0-v1.13.1
infyshine@infyshine:~/judge0-v1.13.1$ sudo docker compose down -v
[+] down 5/5
✓ Container judge0-server Removed 10.1s
✓ Container judge0-workers Removed 1.2s
✓ Container judge0-db Removed 0.1s
✓ Container judge0-redis Removed 0.2s
✓ Network judge0-v1131_default Removed 0.1s
infyshine@infyshine:~/judge0-v1.13.1$ sudo docker compose up -d
[+] up 5/5
✓ Network judge0-v1131_default Created 0.0s
✓ Container judge0-server Started 0.3s
✓ Container judge0-workers Started 0.3s
✓ Container judge0-db Started 0.3s
✓ Container judge0-redis Started 0.3s
infyshine@infyshine:~/judge0-v1.13.1$ sudo docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS      PORTS      NAMES
770dda401727   postgres:16.2 "docker-entrypoint.s..." 11 seconds ago Up 11 seconds 5432/tcp
judge0-db
9e1adee693f5   redis:7.2.4 "docker-entrypoint.s..." 11 seconds ago Up 11 seconds 6379/tcp
judge0-redis
92edabdf3fd5   judge0/judge0:1.13.1 "/api/docker-entryp..." 11 seconds ago Up 11 seconds
2358/tcp      judge0-workers
ca1ab649bdda   judge0/judge0:1.13.1 "/api/docker-entryp..." 11 seconds ago Up 11 seconds
0.0.0.0:2358->2358/tcp, [::]:2358->2358/tcp judge0-server
infyshine@infyshine:~/judge0-v1.13.1$ curl -X POST http://localhost:2358/submissions/?wait=true \
-H "Content-Type: application/json" \
-d '{
  "source_code": "print(\"Hello\")",
  "language_id": 71
}'
{"stdout": "Hello\n", "time": "0.013", "memory": 21680, "stderr": null, "token": "f43f7c05-8d39-4610-9453-a
a17e073c16f", "compile_output": null, "message": null, "status": {"id": 3, "description": "Accepted"}}
infyshine@infyshine:~/judge0-v1.13.1$
```

■ Success! Judge0 is now working correctly.

Your final test returned:

```
{
  "stdout":"Hello\n",
  "time":"0.013",
  "memory":21680,
  "status":{
    "id":3,
    "description":"Accepted"
  }
}
```

This means:

- PostgreSQL is working
- Redis is working
- Judge0 Server is working
- Judge0 Workers are working
- Isolate Sandbox is working
- Python code execution is working

What Was the Root Cause?

Initially:

```
stat -fc %T /sys/fs/cgroup
```

returned:

cgroup2fs

which means cgroup v2.

Judge0 v1.13.1 requires cgroup v1 memory controller and was failing with:

Cannot write /sys/fs/cgroup/memory/box-1/tasks
What Fixed It?

You added to GRUB:

```
systemd.unified_cgroup_hierarchy=0
cgroup_enable=memory
cgroup_memory=1
swapaccount=1
```

and after reboot:

```
stat -fc %T /sys/fs/cgroup
```

returned:

tmpfs

which means cgroup v1 is active.

After restarting Judge0:

```
sudo docker compose down -v
sudo docker compose up -d
```

the sandbox started working.

Make Docker Work Without sudo

Currently:

```
docker ps
```

fails.

Run:

```
sudo usermod -aG docker $USER
```

Then:

```
newgrp docker
```

or logout/login.

Verify:

```
docker ps
```

works without sudo.

Test Additional Languages

Python:

```
curl -X POST http://localhost:2358/submissions/?wait=true \
-H "Content-Type: application/json" \
-d '{
  "source_code": "print(10+20)",
  "language_id": 71
}'
```

Java:

```
curl -X POST http://localhost:2358/submissions/?wait=true \
-H "Content-Type: application/json" \
-d '{
  "source_code": "public class Main { public static void main(String[] args){ System.out.println(\"Hello Java\"); } }",
  "language_id": 62
}'
```

C:

```
curl -X POST http://localhost:2358/submissions/?wait=true \
-H "Content-Type: application/json" \
-d '{
  "source_code": "#include<stdio.h>\nint main(){printf(\"Hello C\");}",
  "language_id": 5
}'
```

```
"language_id":50
```

```
}'
```

Final Working Configuration

Ubuntu 24.04 LTS ■

Kernel 6.17.0-35 ■

Judge0 1.13.1 ■

Docker Compose ■

PostgreSQL 16.2 ■

Redis 7.2.4 ■

cgroup v1 enabled via GRUB ■

Code execution working ■

Your Judge0 installation is now fully operational. ■

Judge0 Dummy Client

Judge0 dummy client can be used to try and test features of Judge0. Data is not validated on client side. Client first creates new submission and then fetches submission status every 1500ms. Client stops fetching submission status after receiving an error or when submission status is not In Queue or Processing.

Note that you need to define API URL. This is the base url of some Judge0 host (e.g. <https://api.judge0.com>). Description of network errors can be found in console of your browser, so check that for more information.

API URL

<http://192.168.88.105:2358>

AUTHENTICATION HEADER

X-Auth-Token

AUTHENTICATION TOKEN

AUTHORIZATION HEADER

X-Auth-User

AUTHORIZATION TOKEN

Source Code null

```
print("HELLO")
```

Language ID

71

null

Number Of Runs

1

null

CPU Time Limit

5

null

CPU Extra Time

1

null

Wall Time Limit

10

null

Memory Limit

128000

null

Stack Limit

64000

null

Max Processes And Or Threads

60

null

Enable Per Process And Thread Time Limit true false null

Enable Per Process And Thread Memory Limit true false null

Max File Size

1024

null

Enable Network true false null

Stdin null

Expected Output null

HELLO

Fields

Wait for submission

Send request with Base64 encoded data

Accept response with Base64 encoded data

Run Stop Clear Log Back To Top

Request/Response Log

[Request 19/06/2026, 12:31:27] POST http://192.168.88.105:2358/submissions

```
{
  "source_code": "print(\"HELLO\")",
  "language_id": "71",
  "number_of_runs": null,
```

```
"stdin": null,
"expected_output": "HELLO",
"cpu_time_limit": null,
"cpu_extra_time": null,
"wall_time_limit": null,
"memory_limit": "128000",
"stack_limit": null,
"max_processes_and_or_threads": null,
"enable_per_process_and_thread_time_limit": null,
"enable_per_process_and_thread_memory_limit": null,
"max_file_size": null,
"enable_network": null
}
[Response 19/06/2026, 12:31:27] 201 Created
{
"token": "5097c09e-d3e7-4c30-ab26-a71aebd1c079"
}
```

```
[Request 19/06/2026, 12:31:29] GET
http://192.168.88.105:2358/submissions/5097c09e-d3e7-4c30-ab26-a71aebd1c079
[Response 19/06/2026, 12:31:29] 200 OK
{
"stdout": "HELLO\n",
"time": "0.009",
"memory": 3504,
"stderr": null,
"token": "5097c09e-d3e7-4c30-ab26-a71aebd1c079",
"compile_output": null,
"message": null,
"status": {
"id": 3,
"description": "Accepted"
}
}
```

[DONE 19/06/2026, 12:31:29]

■ Congratulations! Judge0 is working successfully.

Your dummy client test shows:

Request
print("HELLO")

Language ID:

71 = Python 3
Response
{
"stdout": "HELLO\n",
"time": "0.009",
"memory": 3504,
"status": {
"id": 3,
"description": "Accepted"
}
}


```
}
Meaning
Field Result
API Server ■ Working
PostgreSQL ■ Working
Redis ■ Working
Judge0 Workers ■ Working
Sandbox (Isolate) ■ Working
Python Execution ■ Working
Network Access ■ Working
Status ■ Accepted
Why It Works Now
```

Before:

Cannot write /sys/fs/cgroup/memory/box-1/tasks

After enabling cgroup v1:

```
systemd.unified_cgroup_hierarchy=0
cgroup_enable=memory
cgroup_memory=1
swapaccount=1
```

and rebooting:

```
stat -fc %T /sys/fs/cgroup
```

returned:

```
tmpfs
```

which allowed Judge0's sandbox to create:

```
/box
```

and execute code successfully.

Test More Languages

Java

```
{
"source_code": "public class Main { public static void main(String[] args){ System.out.println(\"Hello Java\"); } }",
"language_id": 62
}
```

C

```
{
"source_code": "#include<stdio.h>\nint main(){printf(\"Hello C\");}",
"language_id": 50
}
```

C++

```
{
"source_code": "#include<iostream>\nusing namespace std;\nint main(){cout<<\"Hello CPP\";}",
"language_id": 54
}
```

Save These Commands for Future

Check cgroup:

```
stat -fc %T /sys/fs/cgroup
```

Expected:

tmpfs

Check containers:

```
sudo docker ps
```

Restart Judge0:

```
cd ~/judge0-v1.13.1
```

```
sudo docker compose down
```

```
sudo docker compose up -d
```

View logs:

```
sudo docker logs judge0-server
```

```
sudo docker logs judge0-workers
```

Final Status

- Ubuntu 24.04 LTS
- Docker Installed
- PostgreSQL 16.2
- Redis 7.2.4
- Judge0 1.13.1
- cgroup v1 Enabled
- Python Execution Working
- Judge0 Dummy Client Working

Your Judge0 Online Compiler/Judge API is now fully operational. ■